Task 1: Launch an EC2 instance and view CloudWatch metrics

1. Log in to the AWS Management Console.
2. Open EC2 → Instances → Launch instances.
3. Enter the instance name, select an eligible Linux AMI and choose an eligible t2.micro instance type if available in your region and account.
4. Configure a key pair and the required network settings, then launch the instance.
5. Open CloudWatch → Metrics → All metrics → EC2 → Per-Instance Metrics.
6. Select your instance ID and display the CPUUtilization metric.

Task 2: Create a CPU utilization alarm

Step 1: Select the metric

Open CloudWatch → Alarms → Create alarm → Select metric → EC2 → Per-Instance Metrics → CPUUtilization.

Step 2: Configure the condition

Set the statistic to Average, period to 5 minutes, condition to Greater than 70, and evaluation periods to 1.

Step 3: Configure email notification

Select an SNS topic or create one, enter your email address if creating a new topic, and confirm the subscription using the email AWS sends you.

Step 4: Finish

Name the alarm EC2-CPU-Above-70, review the settings and create the alarm.

Task 3: Monitor Azure Storage

1. Sign in to the Azure Portal.
2. Search for Storage accounts and create or open a storage account.
3. Select Monitoring → Metrics.
4. Choose a metric such as Transactions or Availability.
5. Set the time range and aggregation, then view the graph.

Task 4: Simulate a CPU spike

On your own Linux EC2 instance, you can use the following commands if the stress utility is installed:

- sudo apt update
- sudo apt install stress -y
- stress --cpu 2 --timeout 360

Then:
1. Open CloudWatch → Metrics → EC2.
2. Select your instance's CPU utilization graph.
3. Refresh the graph and wait for the next metric datapoint.
4. Check the alarm state and your email inbox.

Task 5: AWS CloudWatch vs Azure Monitor

Feature	AWS CloudWatch	Azure Monitor
Virtual machine monitoring	Collects EC2 CPU, network and disk-related metrics.	Collects Azure VM metrics and can collect guest operating system data.
Storage monitoring	Tracks supported AWS storage metrics, such as S3 request metrics and EBS volume performance.	Tracks supported storage account metrics, including transactions and availability.

Alerts	Uses metric alarms and integrates with Amazon SNS.	Uses alert rules and action groups for email, SMS and other supported actions.
Log analysis	CloudWatch Logs supports log collection, searching and analysis.	Log Analytics and Azure Monitor Logs support querying and analysis of collected logs.